AWANTIKA SRIVASTAVA

Associate Data Scientist | Statistical Modeling | Linux

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PROFILE SUMMARY

Data Scientist with 2+ years of experience delivering end-to-end analytics and machine learning solutions across predictive modeling and behavioral risk monitoring systems. Strong expertise in regression, classification, decision trees, and statistical validation using Python and SQL. Experienced in executing analytic projects from problem formulation to deployment, ensuring data quality, and translating analytical insights into actionable business outcomes.

CORE TECHNICAL SKILLS

- **Programming & Data Science:** Python, C++, SQL., Pandas, Numpy, Scikit-learn, SciPy
- **Statistics & Mathematics:** Statistical Modeling, Descriptive Statistics, Hypothesis Testing, Probability, Sampling, A/B Testing, Cross-Validation
- **Machine Learning:** Supervised & Unsupervised Learning, Regression, Classification, Clustering, Random Forest, Decision Trees, SVM, KNN, K-Means, XGBoost, LightGBM, Model Evaluation Metrics (Accuracy, Precision, Recall, F1-score, ROC-AUC).
- **Deep Learning:** ANN, CNN, RNN, LSTM, BERT, Transfer Learning, Self-Attention, multi-head Attention, Time-Series Forecasting, Tensorflow, PyTorch, Keras
- **NLP:** Text preprocessing, Tokenization, Chunking, Sentiment Analysis, Transformer-based models, NLTK, TextBlob, SpaCy
- **Data Mining & Analytics:** Data Cleaning, Aggregation, EDA, Pattern Recognition, Model Performance Metrics
- **Computer Vision:** Image Classification & Preprocessing, Object detection (YOLO, SSD, MobileNet, ResNet), Paddle OCR, Video Analytics (RTSP), OpenCV
- **Data Tools & Reporting:** Excel, PowerPoint, PowerBI
- **Project Execution & Governance:** End-to-End Data Science Lifecycle, Analytical Planning, Data QA & Validation, Documentation, Stakeholder Communication, Business Insight Translation

EXPERIENCE

AI/ML Engineer | PPS International Pvt. Ltd. **January 2024-Present**

- Executed **end-to-end data science projects** from **business requirement gathering** and **problem formulation** to **model deployment** and stakeholder reporting.
- Developed and optimized **regression** and **classification models** (Random Forest, Decision Trees, Logistic Regression) for **predictive analytics** and improved decision-making accuracy.
- Extracted, transformed, and aggregated large structured datasets using **SQL** (joins, subqueries, aggregations) to support modeling and reporting.
- Performed rigorous **statistical validation**, including **cross-validation**, **hypothesis testing**, and performance benchmarking to ensure model robustness.
- Conducted detailed **Exploratory Data Analysis (EDA)** to identify trends, patterns, and key business drivers.
- Implemented **data quality assurance (QA)** checks to maintain **data consistency, completeness, and accuracy** across analytical pipelines.
- Designed effective **feature engineering** strategies to enhance **model interpretability** and predictive performance.
- Collaborated with **cross-functional stakeholders** to define **analytical scope**, align on KPIs, and deliver business-driven insights.
- Created structured **documentation** covering methodologies, assumptions, validation metrics, and model limitations to ensure reproducibility.
- Translated analytical findings into **actionable business recommendations** through executive-level presentations and reporting.
- Promoted **best practices in data mining and statistical Modeling** to support continuous improvement initiatives.

PROJECTS

Railway Driver Assistance System (RDAS) | Enterprise ML Project

- **Designed and deployed a** real-time computer vision–based ML system **for unsafe** driver behavior detection using **CNN-based SSD MobileNet** model.
- Trained and optimized models on large-scale video datasets, achieving **20–25 FPS** real-time processing with **<150 ms inference latency**.
- Implemented **end-to-end ML pipelines** for data ingestion, preprocessing, model training, evaluation, and production inference.
- Deployed optimized models using **TensorFlow Lite** on edge/production environments for continuous monitoring.
- Built a **Flask-based web dashboard** to visualize detections and automatically record **30-second event clips**, reducing manual review effort.

Chatbot Using LLM & RAG | Applied ML Project

- Built a Lightweight **LLM-Powered chatbot** using **TinyLLaMA** to answer user queries over content.
- Implemented a **Retrieval-Augmented Generation (RAG) pipeline** to retrieve relevant resume sections for contextual question answering.
- Selected **TinyLLaMA** to ensure **low memory footprint and fast inference**, making the solution suitable for resource-constrained environments.
- Applied **prompt engineering techniques** to improve response relevance and consistency.
- Deployed the chatbot as an interactive **Streamlit web application** for real-time user interaction.

YouTube Comments Sentiment Analyzer | link-https://youtube-ai-analyzer-ndzqo6r2mepjrtsdjmwaxl.streamlit.app/

- Developed **LSTM-based time-series forecasting** model trained on 5+ years historical stock data reducing **RMSE** by 18**%.**
- Engineered lag **features**, **rolling statistics**, and volatility indicators improving prediction stability.
- **Implemented** custom **BPTT training** loop with adaptive learning rate scheduling for **stable convergence**.
- Applied **regularization techniques** reducing overfitting gap by **20**%.

CERTIFICATION

- IBM Data Science & AI Certification
- AWS Generative AI with Large Language Models
- OpenCV Computer Vision Certification

EDUCATION

IMS Engineering College, Ghaziabad **September - 2020**

Bachelor of Technology (Electrical and electronics engineering)